

FENG HE

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EDUCATION

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- M.S. in Neuroscience, University of Science and Technology of China 2023.09 – 2026.11
- Research focus: machine learning, representation learning, and scientific imaging
 - Recipient of the **China National Scholarship** (Top 0.1% nationwide), 2026
- B.S. in Computer Science, Yangtze University 2019.09 – 2023.06
- GPA: 4.07/5.0; **Ranked 1st** overall among ~300 students (graduate recommendation)

RESEARCH INTERESTS

LLM-based AI agents, including planning, tool use, reliable multi-step reasoning and autonomous decision making. Broad interests in scientific foundation models, multimodal representations and cross-domain learning that support robust agent behavior in real-world tasks

SELECTED PUBLICATIONS

Full publication list available on [Google Scholar](#). Total citations: 467 (as of February 2026).

1. **Feng He**, Guodong Tan, Qiankun Li, Jun Yu, and Quan Wen. From Pixels to Views: Learning Angular-Aware and Physics-Consistent Representations for Light Field Microscopy. *NeurIPS 2025*.
 - Developed physics-consistent 3D representations for biological light-field microscopy using multi-view angular cues.
 2. **Feng He**, Hanlin Li, Xin Ning, and Qiankun Li. BeautyDiffusion: Generative Latent Decomposition for Makeup Transfer via Diffusion Models. *Information Fusion (IF: 18+)*, 2025.
 - Introduced a generative latent decomposition framework enabling disentangled makeup transfer in diffusion models.
 3. **Feng He**, Zhenting Wang, Dongfang Liu, Ruixiang Tang, and Qiankun Li. Productive LLM Hallucinations: Conditions, Mechanisms, and Benefits. *ECCV 2026 Submission*.
 - Analyzed LLM reasoning under controlled semantic perturbations in scientific domains (protein, drug, medical), characterizing entropy shifts, convergence stability, and mechanism-level factors that govern beneficial hallucinations.
 4. Junyuan Mao*, **Feng He***, Kun Wang, Yang Liu, Lingjuan Lyu, Qiankun Li, and Yueming Jin. Hallucination Is Not Always an Error: On the Theory and Mechanisms of Beneficial Hallucinations. *ICML 2026 submission*.
 - Theoretical and mechanistic analysis of beneficial hallucinations in large language models.
 5. Qiankun Li, **Feng He**, Huabao Chen, Xin Ning, Kun Wang, and Zengfu Wang. Unleashing Foundation Vision Models: Adaptive Transfer for Diverse Data-Limited Scientific Domains. *NeurIPS 2025*.
 - Developed a general-purpose fine-tuning adapter enabling foundation vision models to transfer across diverse data-limited scientific domains (biomedical, industrial, agricultural, materials).
 6. Daichao Zhao, Qiupu Chen, **Feng He**, Xin Ning, and Qiankun Li. High-Fidelity Generation of Lane Scenes under Adverse Weather and Lighting Conditions without Re-annotation. *CVPR 2026*.
 - Contributed to manuscript writing, revision and presentation polishing for a lane-scene generation framework under diverse weather and lighting conditions.
- * Equal contribution.

HONORS AND AWARDS

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- Grand Challenge Champion (1st Place)-Micro-Action Recognition ACM MM' 2024
- Merit Student; Hanko Scholarship Yangtze University

INTERN EXPERIENCE

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- Beijing Academy of Artificial Intelligence (BAAI)**, Research Intern 2026.02 – Present
- Conducting AI agent research with a focus on large language model based planning, tool use and autonomous decision making. Developing and evaluating agent architectures that enhance reliability and multi step reasoning.

Rochester Institute of Technology, Research Collaborator

2025.05 – 2026.01

Collaborating with Prof. Dongfang Liu on research related to hallucination-aware learning and reasoning in large language models. A joint paper is currently under review at *ECCV 2026*.

University of Science and Technology of China, Research Assistant

2023.09 – 2025.05

Conducting research on AI-driven biological image analysis and scientific imaging. Developing novel representation learning methods, resulting in first-author publications at *NeurIPS 2025*.